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B.Tech

Big Data

KCS-061

With Notes

UNIT-1

Introduction to Big Data

(in one video)

AKTU Exam

Topics to be covered...

Evolution of Technology
What is Big Data?
Types of Big Data?
Big Data Examples & Use Cases
Big Data architecture
When to use this architecture
5Vs of Big Data
Big Data technology
Big Data importance
Big Data applications
Big Data Analytics
Need for Big Data Analytics
What is Big Data Analytics
Types of Big Data Analytics
Happy Ending!

What is Big Data?

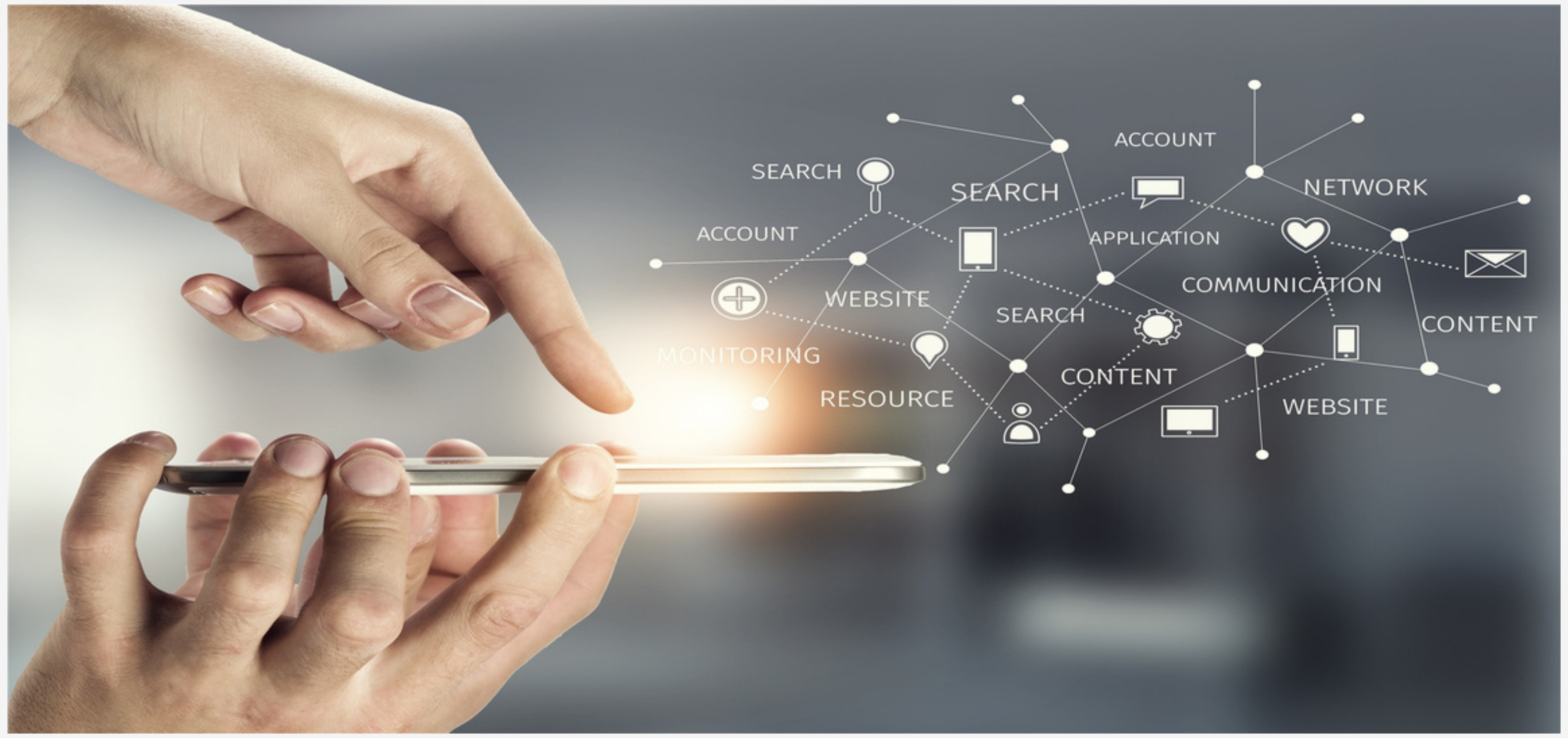
Big data is the term for a collection of data sets so large and complex that it becomes difficult to process using on-hand database management tools or traditional data processing applications

What is Big Data?

Structured
Unstructured
Semi-structured

Structured

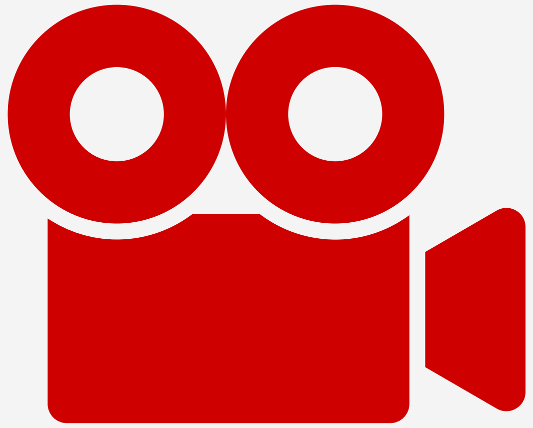
Any data that can be stored, accessed and processed in the form of fixed format is termed as a 'structured' data.



Table

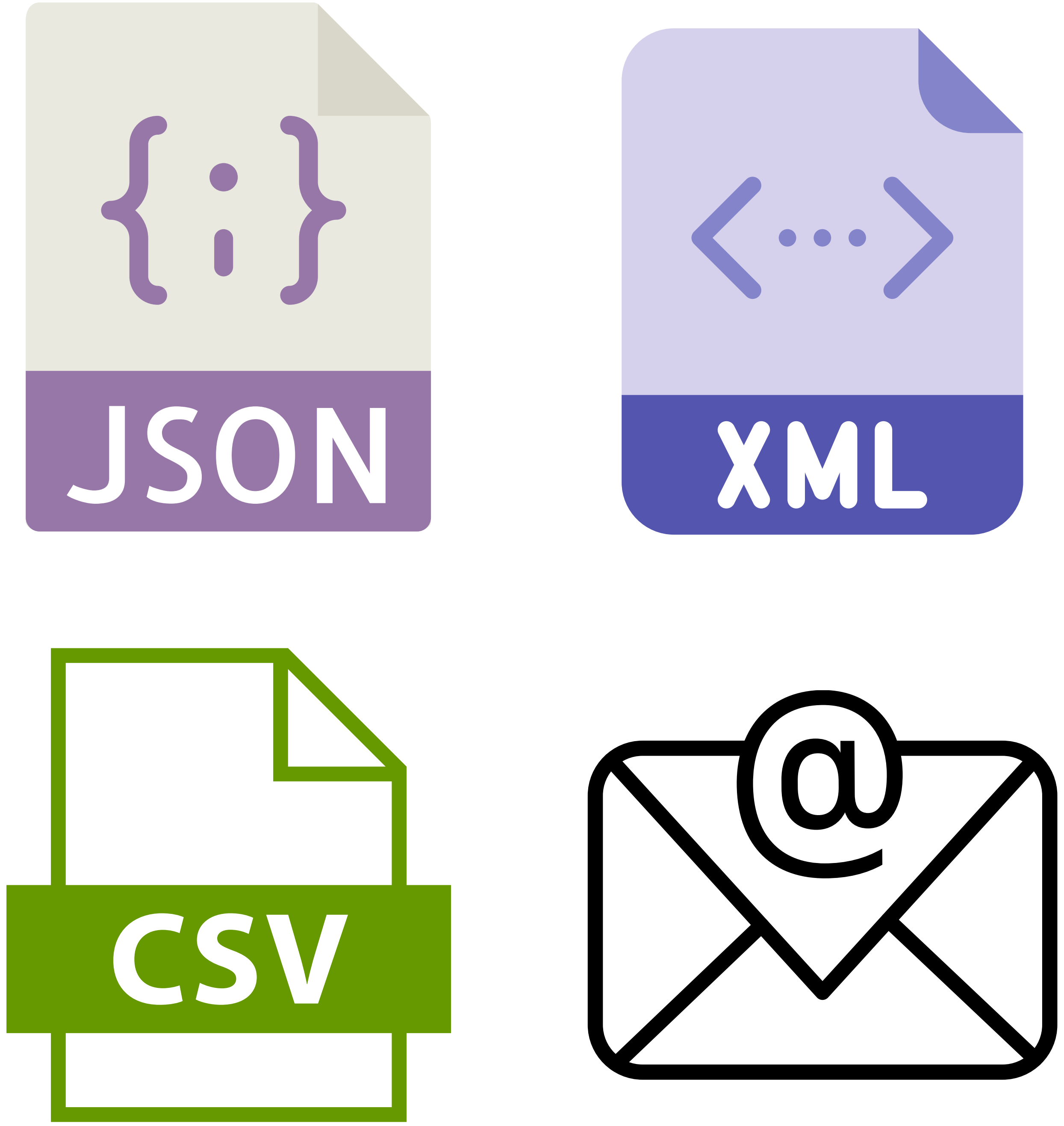
Unstructured

Any data with unknown form or the structure is classified as unstructured data.



Semi-structured

Semi-structured data is information that does not reside in a relational database or any other data table, but nonetheless has some organizational properties to make it easier to analyze, such as semantic tags.



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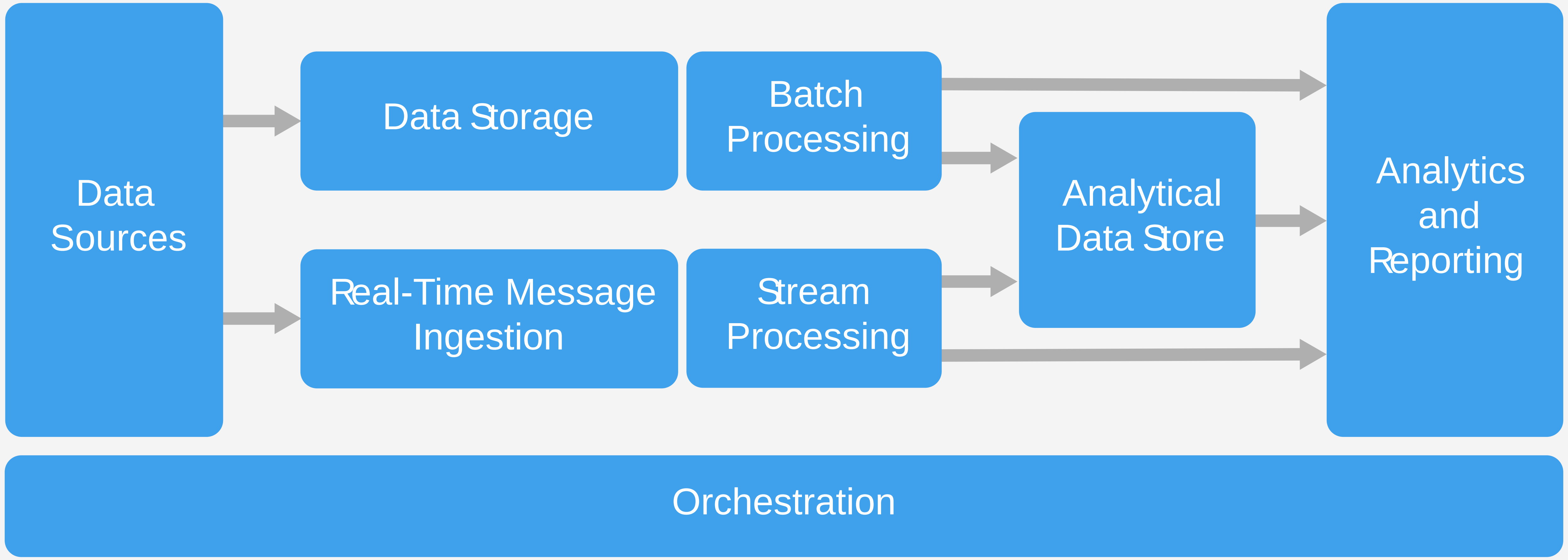
8 Big Data Examples & Use Cases

- Transportation.
- Advertising and Marketing.
- Banking and Financial Services.
- Government.
- Media and Entertainment.
- Meteorology.
- Healthcare.
- Cybersecurity.

Big Data architecture

A big data architecture is designed to handle the ingestion, processing, and analysis of data that is too large or complex for traditional database systems.



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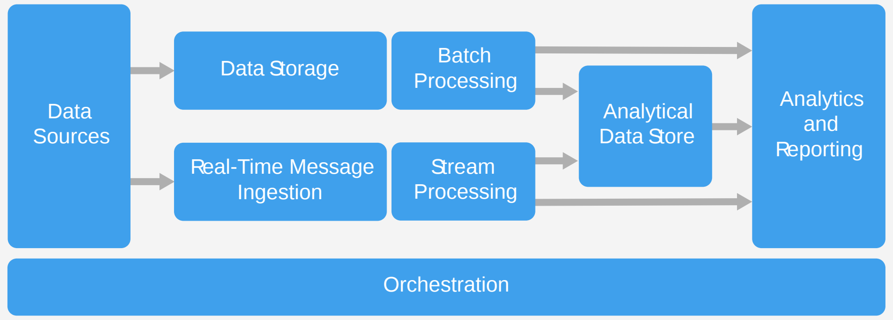
Big Data architecture



Data sources: All big data solutions start with one or more data sources.

Examples include:

-> Application data stores, such as relational databases.

-> Static files produced by applications, such as web server log files.

-> Real-time data sources, such as IoT devices.

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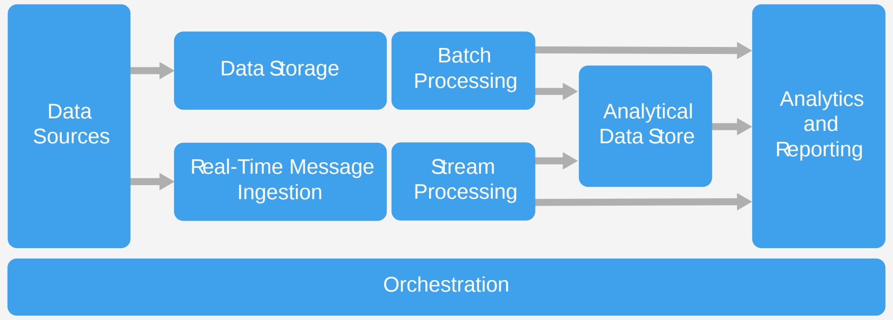
Big Data architecture



Data storage: Data for batch processing operations is typically stored in a distributed file store that can hold high volumes of large files in various formats. This kind of store is often called a data lake. Options for implementing this storage include Azure Data Lake Store or blob containers in Azure Storage.

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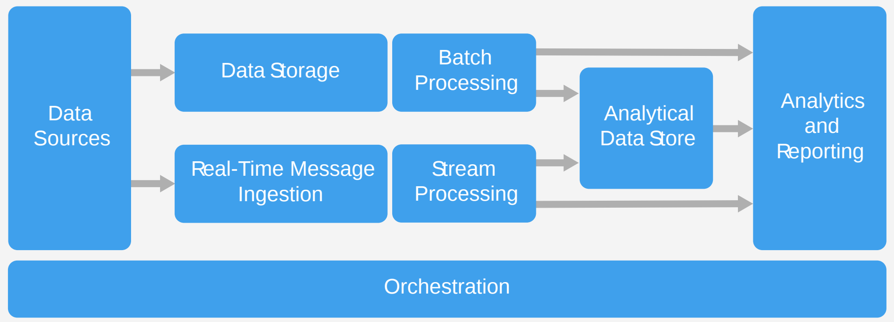
Big Data architecture



Batch processing: Because the data sets are so large, often a big data solution must process data files using long-running batch jobs to filter, aggregate, and otherwise prepare the data for analysis. Usually these jobs involve reading source files, processing them and writing the output to new files. Options include running U-SQL jobs in Azure Data Lake Analytics, using Hive, Pig, or custom Map/Reduce jobs in an HDInsight Hadoop cluster, or using Java, Scala, or Python programs in an HDInsight Spark cluster.

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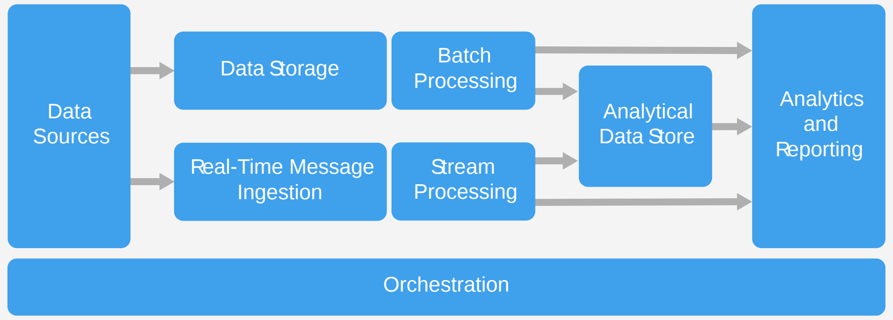
Big Data architecture



Stream processing: After capturing real-time messages, the solution must process them by filtering, aggregating , and otherwise preparing the data for analysis. The processed stream data is then written to an output sink. Azure Stream Analytics provides a managed stream processing service based on perpetually running SQL queries that operate on unbounded streams. You can also use open source Apache streaming technologies like Storm and Spark Streaming in an HDInsight cluster.

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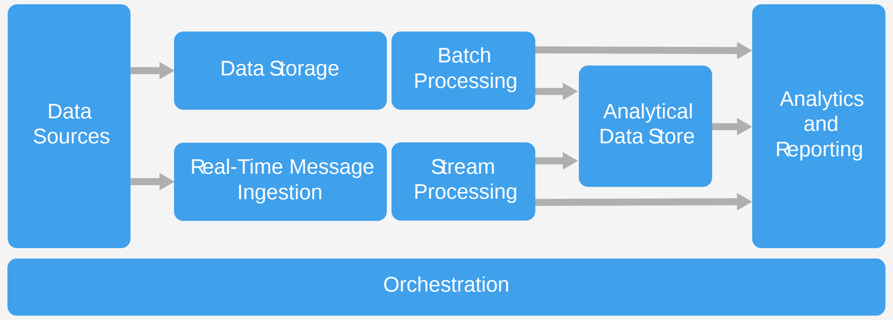
Big Data architecture



Analytical data store: Many big data solutions prepare data for analysis and then serve the processed data in a structured format that can be queried using analytical tools. The analytical data store used to serve these queries can be a Kimball-style relational data warehouse, as seen in most traditional business intelligence (BI) solutions.

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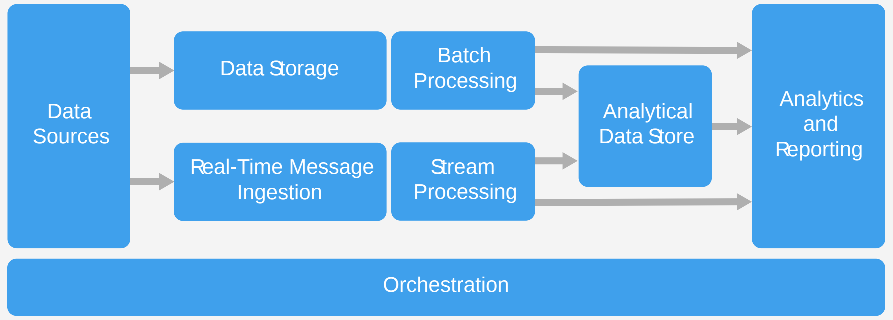
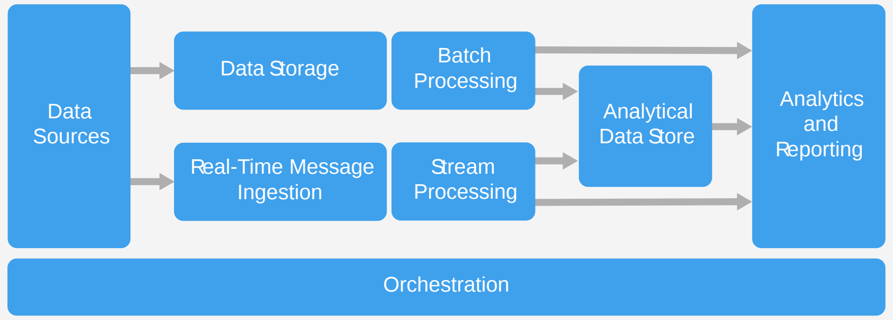
Big Data architecture



Analysis and reporting: The goal of most big data solutions is to provide insights into the data through analysis and reporting. To empower users to analyze the data, the architecture may include a data modelling layer, such as a multidimensional OLAP cube or tabular data model in Azure Analysis Services. It might also support self-service BI, using the modelling and visualization technologies in Microsoft Power BI or Microsoft Excel. Analysis and reporting can also take the form of interactive data exploration by data scientists or data analysts.

Big Data architecture



Orchestration: Most big data solutions consist of repeated data processing operations, encapsulated in workflows, that transform source data, move data between multiple sources and sinks, load the processed data into an analytical data store, or push the results straight to a report or dashboard. To automate these workflows, you can use an orchestration technology such Azure Data Factory or Apache Oozie and Sqoop.

When to use this architecture

1. Store and process data in volumes too large for a traditional database.
2. Transform unstructured data for analysis and reporting.
3. Capture, process, and analyse unbounded streams of data in real time, or with low latency.
4. Use Azure Machine Learning or Microsoft Cognitive Services.

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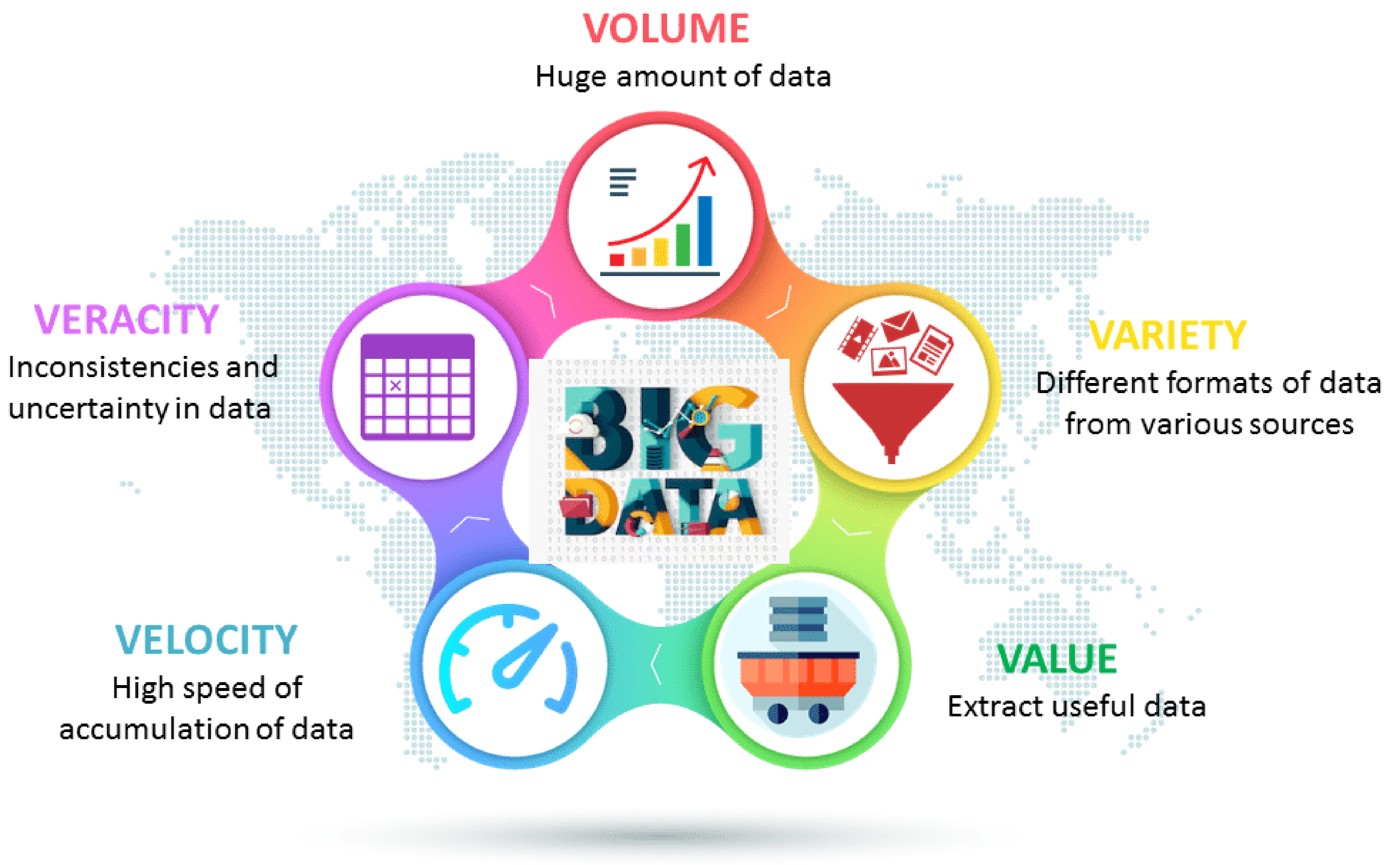
Characteristics OR 5Vs of Big Data

1. Volume
2. Veracity
3. Variety
4. Value
5. Velocity



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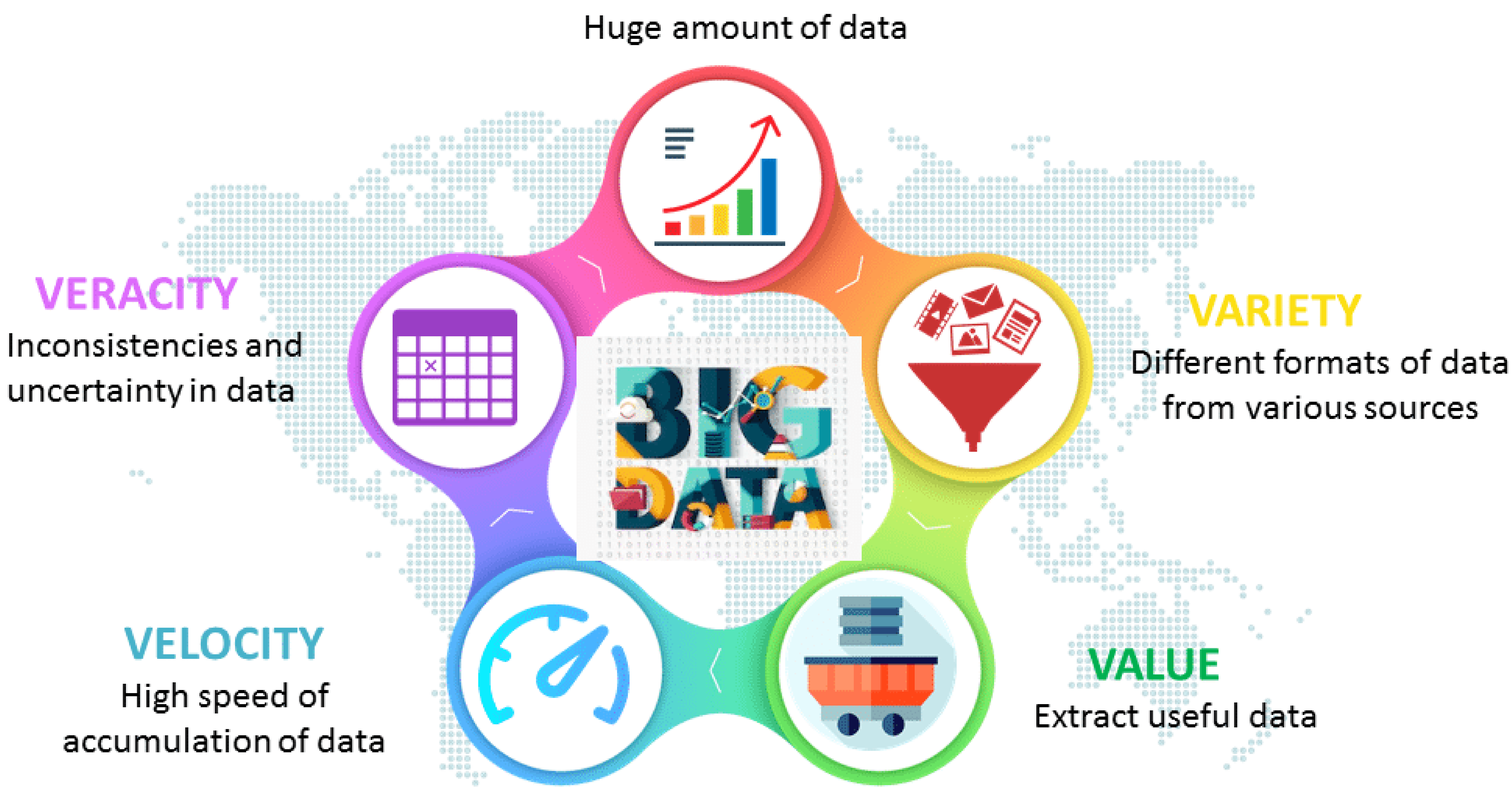

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Big Data technology

Big Data importance

1. Cost Savings
2. Time-Saving
3. Understand the market conditions
4. Social Media Listening
5. Boost Customer Acquisition and Retention
6. Solve Advertisers Problem
7. The driver of Innovations and Product Development

Big Data applications

1. Banking and Securities
2. Communications, Media and Entertainment
3. Healthcare Providers
4. Education
5. Government
6. Insurance
7. Retail and Wholesale trade
8. Transportation

Need for Big Data Analytics

1. Optimize business operations by analyzing customer behaviour

amazon

Need for Big Data Analytics

2. Next Generation Products

NETFLIX

Spotify

What is Big Data Analytics?

Big data analytics is the use of advanced analytic techniques against very large, diverse data sets that include structured, semi-structured and unstructured data, from different sources, and in different sizes from terabytes to zettabytes.

Types of Big Data Analytics

1. Descriptive Analysis
2. Predictive Analysis
3. Prescriptive Analysis
4. Diagnostic Analysis

Types of Big Data Analytics

1. Descriptive Analysis

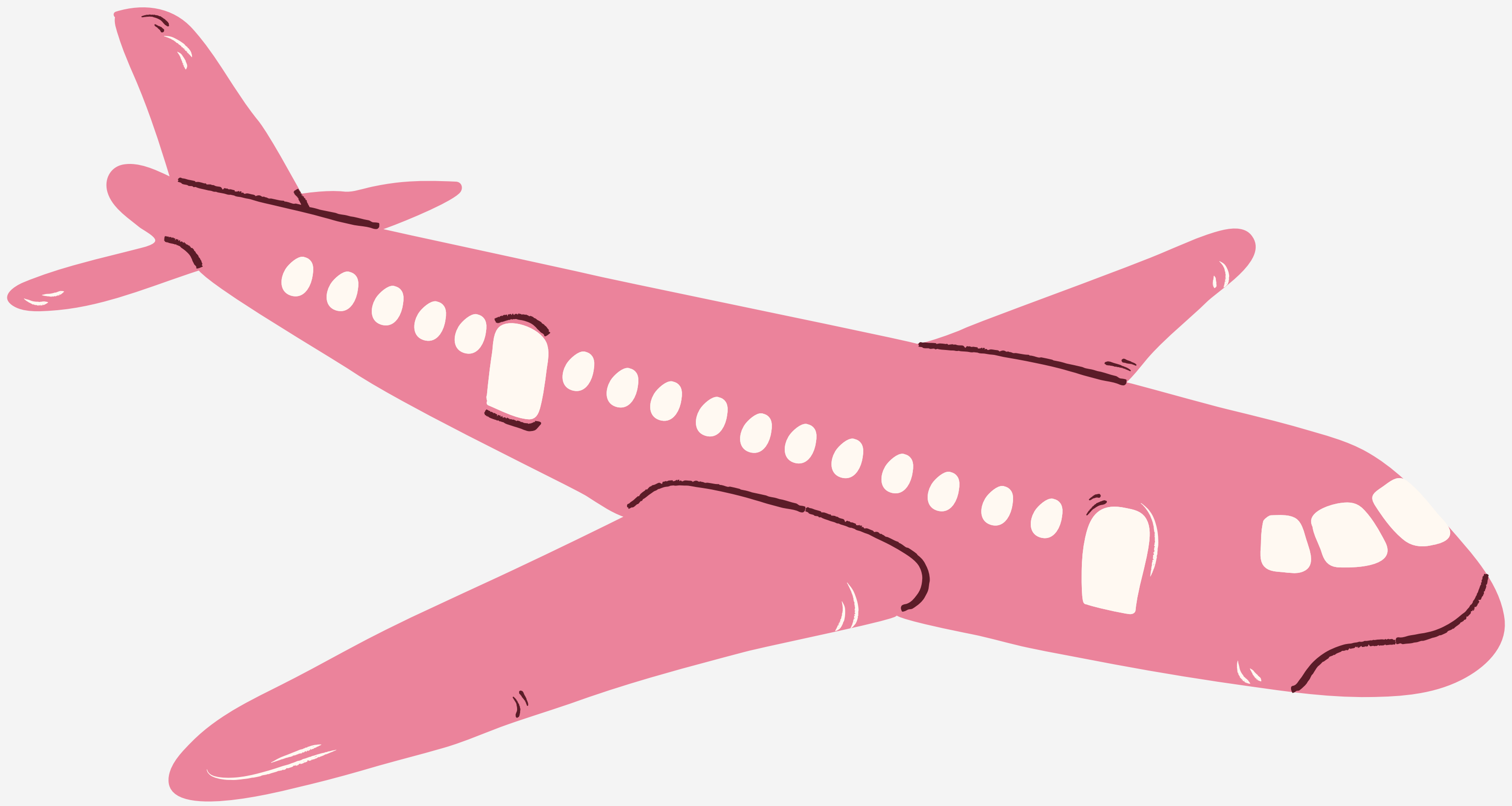
What is happening now based on incoming data.

Google Analytics

Types of Big Data Analytics

2. Predictive Analysis

What might happen in future.



Types of Big Data Analytics

3. Prescriptive Analysis

What action should be taken.

Google's self-driving car is perfect example of Presciptive Analysis.

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Types of Big Data Analytics

4. Diagnostic Analysis

What did it happen